

Block 1: Apex Collections — List, Set, Map

```
List<String> cities = new List<String>{  
    'Hyderabad',  
    'Bhimavaram',  
    'Vijayawada',  
    'Visakhapatnam',  
    'Chennai'  
};  
  
for(String city:cities)  
{  
    System.debug(city.toUpperCase());  
}
```

The screenshot shows the Salesforce Developer Console interface. The top section displays the 'Execution Log' with a table of events. The bottom section shows the 'Logs' table with a list of log entries.

Execution Log

Timestamp	Event	Details
09:53:21:002	USER_DEBUG	[11]DEBUG HYDERABAD
09:53:21:002	USER_DEBUG	[11]DEBUG BHIMAVARAM
09:53:21:002	USER_DEBUG	[11]DEBUG VIJAYAWADA
09:53:21:002	USER_DEBUG	[11]DEBUG VISAKHAPATNAM
09:53:21:002	USER_DEBUG	[11]DEBUG CHENNAI

Logs

User	Application	Operation	Time	Status	Read	Size
ADDALA ANUSHA BHA...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:53:21	Success		4.8 KB
ADDALA ANUSHA BHA...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:50:28	Success		3.89 KB
ADDALA ANUSHA BHA...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:50:07	Success		3.89 KB
ADDALA ANUSHA BHA...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:49:14	Success		2.32 KB
ADDALA ANUSHA BHA...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:48:43	Success		1.86 KB

```

Set<Id> bid=new Set<Id>();

for(Book__c book: [SELECT id,Name FROM Book__c]){

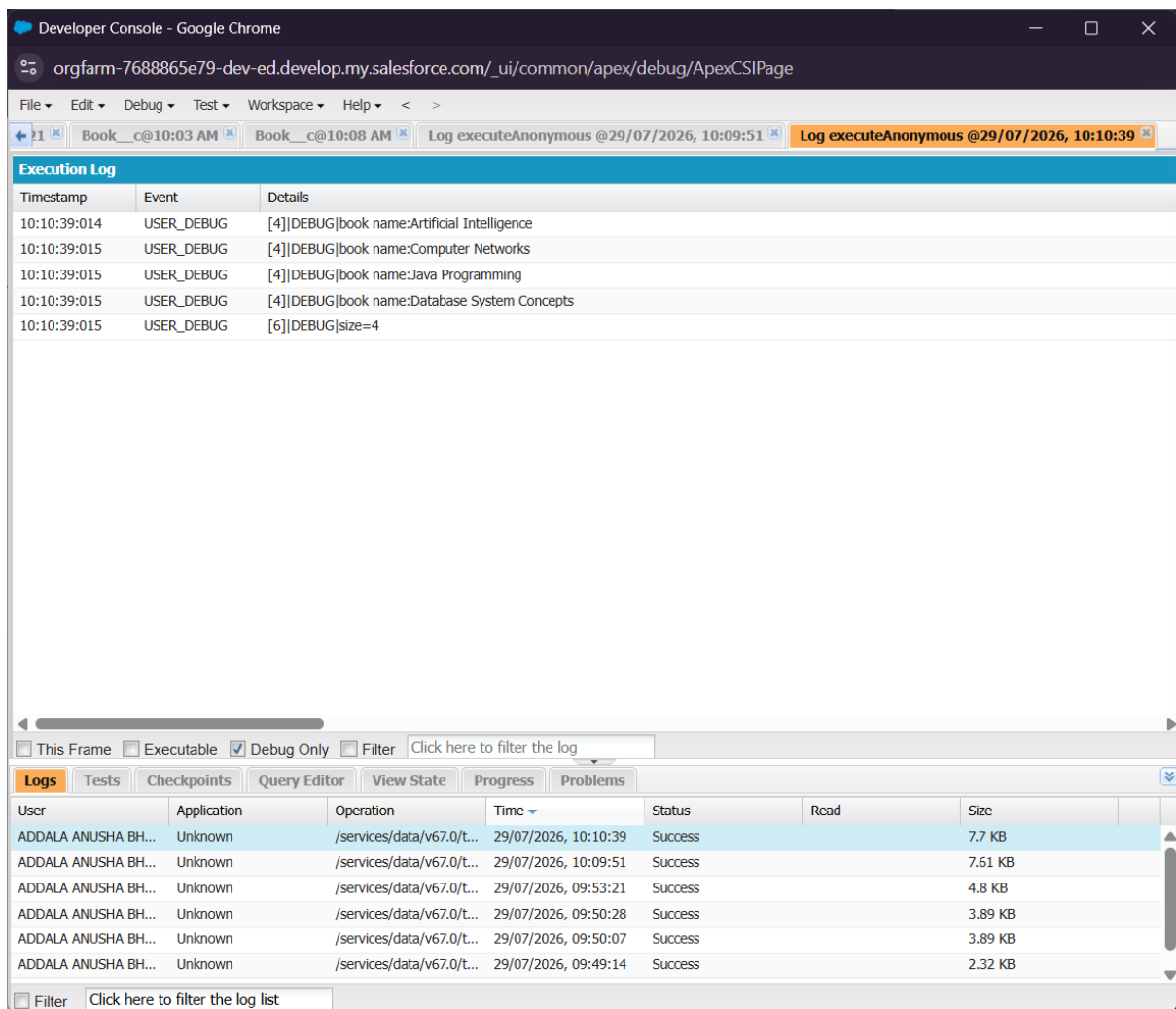
    bid.add(book.id);

    System.debug('book name:' +book.Name);

}

System.debug('size='+bid.size());

```



The screenshot shows the Salesforce Developer Console interface. The top part displays the 'Execution Log' with a table of events. The bottom part shows the 'Logs' tab with a table of log entries.

Execution Log

Timestamp	Event	Details
10:10:39:014	USER_DEBUG	[4] DEBUG book name:Artificial Intelligence
10:10:39:015	USER_DEBUG	[4] DEBUG book name:Computer Networks
10:10:39:015	USER_DEBUG	[4] DEBUG book name:Java Programming
10:10:39:015	USER_DEBUG	[4] DEBUG book name:Database System Concepts
10:10:39:015	USER_DEBUG	[6] DEBUG size=4

Logs

User	Application	Operation	Time	Status	Read	Size
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 10:10:39	Success		7.7 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 10:09:51	Success		7.61 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:53:21	Success		4.8 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:50:28	Success		3.89 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:50:07	Success		3.89 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:49:14	Success		2.32 KB

```

Map<Id, Book__c> bookMap = new Map<Id, Book__c>(
    [SELECT Id, Name, Author__c
    FROM Book__c]
);

```

```
Id bookId = bookMap.keySet().iterator().next();
```

```
Book__c book = bookMap.get(bookId);
```

```
System.debug('Book Id: ' + bookId);
```

```
System.debug('Book Name: ' + book.Name);
```

```
System.debug('Author: ' + book.Author__c);
```

The screenshot displays the Salesforce Developer Console interface. At the top, the browser address bar shows the URL: `orgfarm-7688865e79-dev-ed.develop.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage`. Below the address bar, there are three tabs for log entries: `Log executeAnonymous @29/07/2026, 10:09:51`, `Log executeAnonymous @29/07/2026, 10:10:39`, and `Log executeAnonymous @29/07/2026, 10:13:38`. The main area is titled "Execution Log" and contains a table with the following data:

Timestamp	Event	Details
10:13:38:017	USER_DEBUG	[10][DEBUG]Book Id: a02fj00000msx3XAAQ
10:13:38:018	USER_DEBUG	[11][DEBUG]Book Name: Artificial Intelligence
10:13:38:018	USER_DEBUG	[12][DEBUG]Author: Stuart Russell

Below the Execution Log, there is a section for "Logs" with a table showing various log entries. The table has columns for User, Application, Operation, Time, Status, Read, and Size. The data is as follows:

User	Application	Operation	Time	Status	Read	Size
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 10:13:38	Success		4.88 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 10:10:39	Success		7.7 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 10:09:51	Success		7.61 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:53:21	Success		4.8 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:50:28	Success		3.89 KB
ADDALA ANUSHA BH...	Unknown	/services/data/v67.0/t...	29/07/2026, 09:50:07	Success		3.89 KB

Block 2: Governor Limits & Bulkification

SOQL inside a for loop

```
trigger BookBadTrigger on Book__c (before insert) {
```

```
    for(Book__c b : Trigger.new){
```

```
        List<Book__c> books = [  
            SELECT Id, Name  
            FROM Book__c  
        ];
```

```
        System.debug('Books Count = ' + books.size());
```

```
    }
```

```
}
```

Developer Console - Google Chrome

orgfarm-7688865e79-dev-ed.develop.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

File Edit Debug Test Workspace Help

Log executeAnonymous @29/07/2026, 10:10:39 Log executeAnonymous @29/07/2026, 10:13:38 Log executeAnonymous @29/07/2026, 10:28:31

Execution Log

Timestamp	Event	Details

Execute Anonymous Error

Line: 5, Column: 1
System.LimitException: Too many SOQL queries: 101

OK

☐ This Frame ☐ Executable ☒ Debug Only ☐ Filter Click here to filter the log

Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
ADDALA ANUSHA B...	Unknown	/services/data/v67.0...	29/07/2026, 10:31:42	Too many SOQL qu...	Unread	204.93 KB
ADDALA ANUSHA B...	Browser	/setup/build/editAp...	29/07/2026, 10:31:25	Success	Unread	1.16 KB
ADDALA ANUSHA B...	Unknown	/services/data/v67.0...	29/07/2026, 10:28:31	Success		92.51 KB
ADDALA ANUSHA B...	Unknown	/services/data/v67.0...	29/07/2026, 10:13:38	Success		4.88 KB
ADDALA ANUSHA B...	Unknown	/services/data/v67.0...	29/07/2026, 10:10:39	Success		7.7 KB
ADDALA ANUSHA B...	Unknown	/services/data/v67.0...	29/07/2026, 10:09:51	Success		7.61 KB

☐ Filter Click here to filter the log list

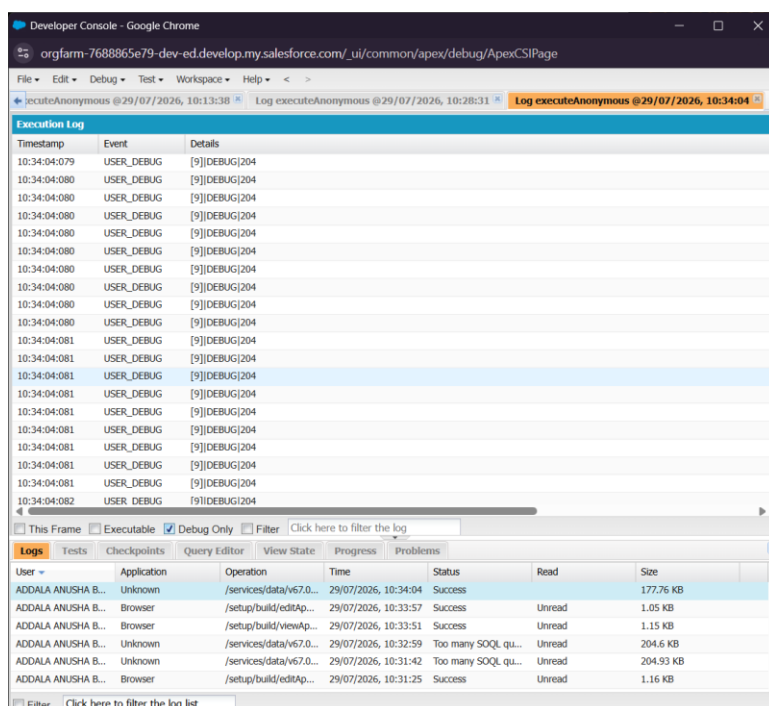
SOQL outside the loop using a List

```
trigger BookBadTrigger on Book__c (before insert) {
```

```
List<Book__c> existingBooks = [
    SELECT Id, Name
    FROM Book__c
];

for(Book__c b : Trigger.new){
    System.debug(existingBooks.size());
}

}
```



Block 3: Asynchronous Apex

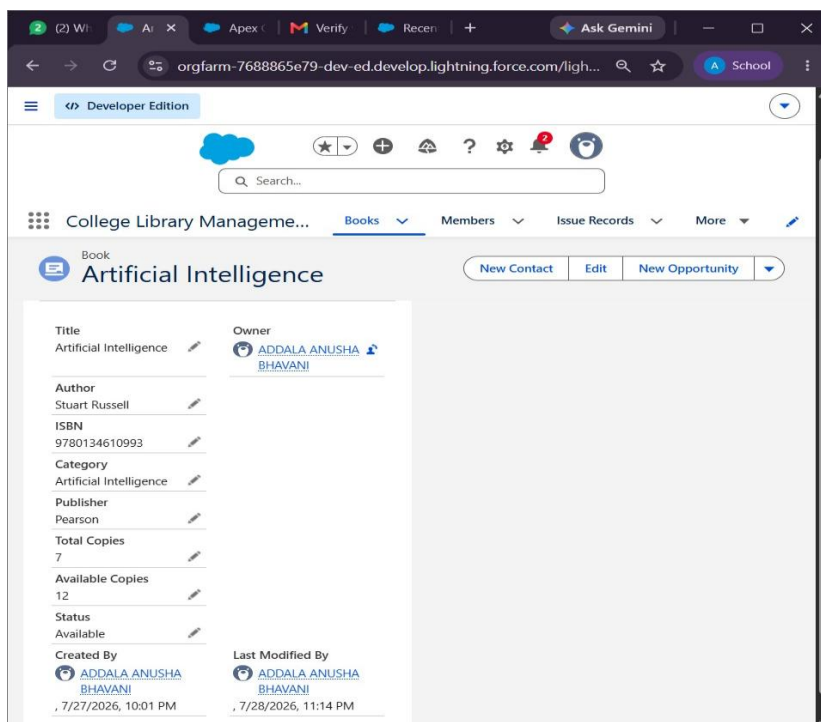
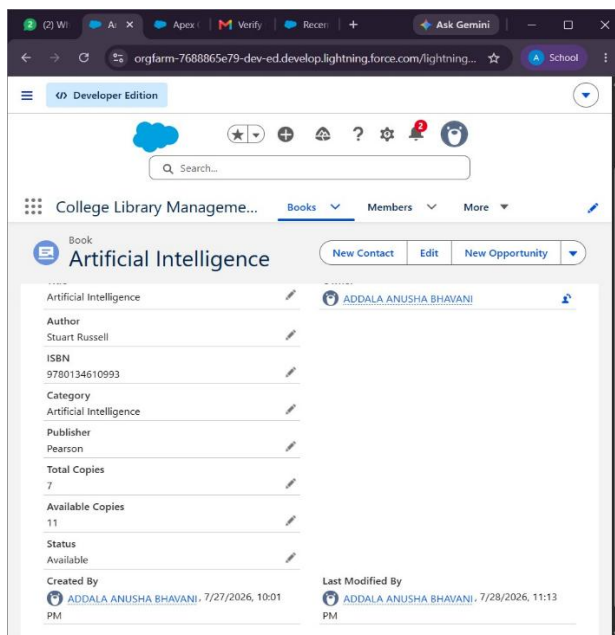
@future method

Before Execution

Available Copies = 11

After Execution

Available Copies = 12



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orgfarm-768865e79-dev-ed:develop.my.salesforce-setup.com/lightning/setup/AsyncApexJobs/home

Developer Edition

Search Setup

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apex jobs

Environments

Jobs

Apex Jobs

Didn't find what you're looking for? Try using Global Search.

Click here to go to the new batch jobs page

Apex Jobs

Monitor the status of all Apex jobs, and optionally, about jobs that are in progress.

Percent of Asynchronous Apex Used: 0%
You have currently used 1 asynchronous Apex operations out of an allowed 24-hour organization limit of 250,000. To learn about how this limit is calculated and what contributes to it, see the [Lightning Platform Apex Limits](#) topic.

View: [All v] [Create New View](#)

Action	Submitted Date	Job Type	Status	Status Detail	Total Batches	Batches Processed	Failures	Submitted By	Completion Date	Apex Class	Apex Method	Apex Job ID
	7/28/2026, 10:29 PM	Future	Completed		0	0	0	ANUSHA BHAVANI AGGALA	7/28/2026, 10:29 PM	BookFutureHandler	updateBook	707f00000uJp

Comp: bootchold
Lambert bootchold

Future

Future Apex is an asynchronous Apex method that runs in the background after the current transaction completes.

Mainly Used for:

- ↳ Calls to external APIs.
- ↳ Long-running operations.
- ↳ Processing tasks without making the user wait.

Syntax:

```
public static void sendEmail(String email) {
```

Queueable Apex

Queueable Apex is an asynchronous job that is more powerful than future Apex.

- ↳ It allows passing complex objects.
- chaining one Queueable job to another
- Monitoring job status.

Syntax:

```
System.enqueueJob(new MyQueueableClass());
```

Batch Apex

Batch Apex is used to process a very large number of records by dividing them into smaller batches. Each batch runs independently and separately. It helps to avoid governor limits.

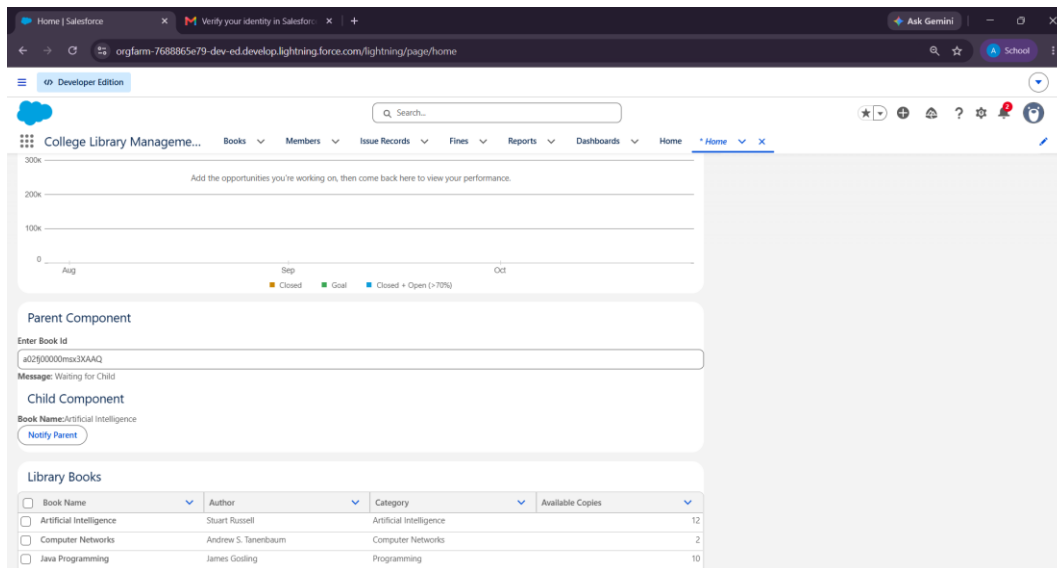
Syntax:

```
Database.executeBatch(new MyBatchClass());
```

Block 4: LWC — Component Communication

https://drive.google.com/file/d/1f5ETre_qZISofqPUVpFGM12l1l4gi0Dy/view?usp=sharing

Book Name	Author	Category	Available Copies
☐ Artificial Intelligence	Stuart Russell	Artificial Intelligence	12
☐ Computer Networks	Andrew S. Tanenbaum	Computer Networks	2
☐ Java Programming	James Gosling	Programming	10
☐ Database System Concepts	Abraham Silberschatz	Database	5
☐ Book 1			
☐ Book 2			



Today I learned how to use asynchronous Apex (@future, Queueable, and Batch Apex) and how to build parent-to-child and child-to-parent communication in Lightning Web Components using @api properties and custom events. I'm still stuck on understanding when to choose Future, Queueable, or Batch Apex in real-world scenarios without referring to examples.